Online Bus Ticket Booking System

Aim

To design and develop a mobile-responsive bus ticket booking website in which users can search buses, select seats, book and cancel tickets and post comments, and the admin can manage bus categories, bus details, users and reports.

Algorithm

1. Start the application; the user registers and logs in (with change password and forgot password options).
2. User searches buses by route, date and time and views the available seats.
3. User selects a bus and seats and enters the passenger details.
4. User pays through the payment gateway; the system saves the ticket and sends a confirmation email.
5. User views the booking history, cancels a ticket if required (the seat is released) and posts comments on a bus.
6. Admin logs in and adds bus categories (Volvo, sleeper coach, coach) and adds or updates bus details such as type, photo, description, route and seats.
7. Admin manages registered users, comments and news.
8. Admin generates the booking report and logs out.

CODE:

HTML:

```
<!DOCTYPE html>
<html lang="en">
<head>
  <meta charset="UTF-8">
  <meta name="viewport" content="width=device-width, initial-scale=1.0">
  <title>BusGo - Online Bus Booking</title>
  <link rel="stylesheet" href="Bus Ticket.css">
</head>
<body>

<header>
  <h1>🚌 BusGo</h1>
  <nav>
    <a href="#home">Home</a>
```

```
      <a href="#search">Search Buses</a>
      <a href="#booking">Book Ticket</a>
      <a href="#history">My Tickets</a>
      <a href="#account">Login/Register</a>
      <a href="#comments">Comments</a>
      <a href="#admin">Admin</a>
    </nav>
</header>

<section class="hero" id="home">
    <h2>Your Journey Starts Here!</h2>
    <p>Book your bus tickets easily, quickly and comfortably.</p>
    <a class="button" href="#search">Find Your Bus</a>
</section>

<section id="search">
    <h2>Search Buses</h2>

    <form id="searchForm">
      <label>From</label>
      <input type="text" id="from" placeholder="Departure city" required>

      <label>To</label>
      <input type="text" id="to" placeholder="Destination city" required>

      <label>Journey Date</label>
      <input type="date" id="journeyDate" required>

      <button type="submit">Search Buses</button>
    </form>

    <div id="searchResults" class="bus-container"></div>
</section>

<section id="booking">
    <h2>Book Your Ticket</h2>
```

```
    <form id="bookingForm">
      <label>Passenger Name</label>
      <input type="text" id="passengerName" required>

      <label>Email</label>
      <input type="email" id="passengerEmail" required>

      <label>Select Bus</label>
      <select id="busSelect" required>
        <option value="">Choose a bus</option>
      </select>

      <label>Journey Date</label>
      <input type="date" id="bookDate" required>

      <label>Select Seat</label>
      <select id="seatSelect" required>
        <option value="">Select a seat</option>
      </select>

      <p id="seatMessage">Choose a bus and date to see available seats.</p>

      <label>Passenger Age</label>
      <input type="number" id="passengerAge" min="1" max="120" required>

      <button type="submit">Confirm Booking</button>
    </form>

    <p id="bookingMessage"></p>
  </section>

  <section id="history">
    <h2>My Ticket History</h2>
    <div id="ticketHistory"></div>
  </section>
```

```
<section id="account">
  <h2>User Account</h2>

  <div class="form-container">
    <form id="registerForm">
      <h3>Create Account</h3>

      <label>Full Name</label>
      <input type="text" id="regName" required>

      <label>Email</label>
      <input type="email" id="regEmail" required>

      <label>Password</label>
      <input type="password" id="regPassword" required>

      <button type="submit">Register</button>
    </form>

    <form id="loginForm">
      <h3>Login</h3>

      <label>Email</label>
      <input type="email" id="loginEmail" required>

      <label>Password</label>
      <input type="password" id="loginPassword" required>

      <button type="submit">Login</button>
      <button type="button" onclick="forgotPassword()">Forgot Password?</button>
    </form>
  </div>

  <button onclick="logout()">Logout</button>
  <p id="accountMessage"></p>
```

```
</section>

<section id="comments">
  <h2>Passenger Comments</h2>

  <form id="commentForm">
    <label>Your Name</label>
    <input type="text" id="commentName" required>

    <label>Select Bus</label>
    <select id="commentBus" required>
      <option value="">Choose a bus</option>
    </select>

    <label>Comment</label>
    <textarea id="commentText" rows="4" required></textarea>

    <button type="submit">Post Comment</button>
  </form>

  <div id="commentList"></div>
</section>

<section id="admin">
  <h2>Administrator Panel</h2>

  <p>Demo admin email: admin@busgo.com</p>
  <p>Demo password: admin123</p>

  <form id="adminLoginForm">
    <label>Admin Email</label>
    <input type="email" id="adminEmail" required>

    <label>Password</label>
    <input type="password" id="adminPassword" required>
```

```
    <button type="submit">Admin Login</button>
</form>

<div id="adminPanel" hidden>
    <h3>Add or Update Bus</h3>

    <form id="busForm">
        <label>Bus Name</label>
        <input type="text" id="busName" required>

        <label>Bus Category</label>
        <select id="busCategory">
            <option>Volvo AC</option>
            <option>Sleeper Coach</option>
            <option>Seater Coach</option>
            <option>Non-AC Bus</option>
        </select>

        <label>From</label>
        <input type="text" id="busFrom" required>

        <label>To</label>
        <input type="text" id="busTo" required>

        <label>Departure Time</label>
        <input type="time" id="busTime" required>

        <label>Ticket Price (₹)</label>
        <input type="number" id="busPrice" min="1" required>

        <label>Total Seats</label>
        <input type="number" id="busSeats" min="1" max="60" required>

        <label>Bus Photo URL (optional)</label>
        <input type="url" id="busPhoto" placeholder="https://example.com/bus.jpg">
```

```
            <label>Description</label>
            <textarea id="busDescription" rows="3"></textarea>

            <button type="submit">Save Bus</button>
        </form>

        <h3>Manage Bus Information</h3>
        <div id="adminBusList"></div>

        <h3>Manage Registered Users</h3>
        <div id="adminUserList"></div>

        <h3>Manage Bookings</h3>
        <div id="adminTicketList"></div>

        <h3>Reports</h3>
        <button onclick="generateReport()">Generate Booking Report</button>
        <div id="reportOutput"></div>

        <h3>News and Announcements</h3>
        <textarea id="newsText" rows="3" placeholder="Enter announcement"></textarea>
        <button onclick="saveNews()">Publish News</button>
        <div id="newsOutput"></div>
    </div>
</section>

<footer>
    <p>© 2026 BusGo Online Ticket Booking System</p>
    <p>Travel safely, travel comfortably!</p>
</footer>

<script src="script.js"></script>
</body>
</html>
```

CSS:

```
* {
    margin: 0;
    padding: 0;
    box-sizing: border-box;
}

html {
    scroll-behavior: smooth;
}

body {
    font-family: Arial, sans-serif;
    background: #f3f6ff;
    color: #29284f;
    line-height: 1.6;
}

header {
    background: linear-gradient(90deg, #172554, #2563eb);
    color: white;
    padding: 20px 5%;
    display: flex;
    justify-content: space-between;
    align-items: center;
    flex-wrap: wrap;
    gap: 15px;
}

header h1 {
    font-size: 29px;
}

nav {
    display: flex;
    flex-wrap: wrap;
    gap: 6px;
```

```
}

nav a {
    color: white;
    text-decoration: none;
    padding: 9px 12px;
    border-radius: 8px;
}

nav a:hover {
    background: #60a5fa;
}

section {
    padding: 35px 6%;
    scroll-margin-top: 15px;
}

section h2 {
    color: #1d4ed8;
    margin-bottom: 20px;
    border-left: 5px solid #f97316;
    padding-left: 12px;
}

.hero {
    margin: 25px 5%;
    padding: 55px 6%;
    border-radius: 22px;
    background: linear-gradient(120deg, #bfdbfe, #dbeafe, #ffedd5);
    box-shadow: 0 8px 25px #17255415;
}

.hero h2 {
    font-size: 38px;
    border: none;
```

```
    padding: 0;
}

.hero p {
    font-size: 18px;
    margin-bottom: 20px;
}

.button {
    display: inline-block;
    background: #f97316;
    color: white;
    text-decoration: none;
    padding: 12px 22px;
    border-radius: 9px;
    font-weight: bold;
}

.button:hover {
    background: #c2410c;
}

form {
    background: white;
    max-width: 650px;
    padding: 23px;
    margin: 15px 0;
    border-radius: 16px;
    box-shadow: 0 5px 20px #17255412;
}

label {
    display: block;
    font-weight: bold;
    color: #1d4ed8;
    margin: 12px 0 5px;
```

```
}

input, select, textarea {
    width: 100%;
    padding: 11px;
    border: 1px solid #cbd5e1;
    border-radius: 8px;
    background: #fbfdff;
    font-size: 15px;
}

input:focus, select:focus, textarea:focus {
    outline: 2px solid #93c5fd;
}

button {
    background: linear-gradient(90deg, #2563eb, #4f46e5);
    color: white;
    border: none;
    padding: 11px 17px;
    margin: 10px 4px 0 0;
    border-radius: 8px;
    cursor: pointer;
    font-size: 14px;
}

button:hover {
    background: #1e40af;
}

.bus-container {
    display: grid;
    grid-template-columns: repeat(auto-fit, minmax(240px, 1fr));
    gap: 20px;
    margin-top: 20px;
}
```

```
.bus-card {
   background: white;
   padding: 20px;
   border-radius: 16px;
   border-top: 5px solid #f97316;
   box-shadow: 0 6px 18px #17255415;
   transition: 0.3s;
   overflow-wrap: anywhere;
}

.bus-card:hover {
   transform: translateY(-5px);
}

.bus-card h3 {
   color: #1d4ed8;
   margin-bottom: 8px;
}

.bus-card img {
   width: 100%;
   max-height: 150px;
   object-fit: cover;
   border-radius: 10px;
   margin-bottom: 12px;
}

.bus-price {
   font-size: 20px;
   color: #ea580c;
   font-weight: bold;
}

.form-container {
   display: grid;
```

```
    grid-template-columns: repeat(auto-fit, minmax(250px, 1fr));
    gap: 20px;
}

#ticketHistory > div,
#commentList > div,
#adminBusList > div,
#adminUserList > div,
#adminTicketList > div,
#newsOutput > div {
    background: white;
    border-radius: 12px;
    padding: 18px;
    margin: 12px 0;
    border-left: 5px solid #2563eb;
    box-shadow: 0 4px 15px #17255410;
    overflow-wrap: anywhere;
}

#adminPanel {
    background: #e7efff;
    padding: 25px;
    border-radius: 18px;
    margin-top: 25px;
}

#adminPanel[hidden] {
    display: none;
}

#reportOutput {
    background: white;
    padding: 18px;
    margin-top: 15px;
    border-radius: 12px;
    overflow-wrap: anywhere;
```

```
}

footer {
  background: #172554;
  color: white;
  text-align: center;
  padding: 25px;
  margin-top: 30px;
}

@media (max-width: 700px) {
  header {
    flex-direction: column;
    align-items: flex-start;
  }

  nav a {
    font-size: 13px;
    padding: 7px;
  }

  .hero h2 {
    font-size: 28px;
  }

  section {
    padding: 25px 5%;
  }
}
```

JAVASCRIPT:

```
const defaultBuses = [
  {
    id: 1,
    name: "Express Volvo",
    category: "Volvo AC",
```

```
    from: "Chennai",
    to: "Bangalore",
    time: "08:00",
    price: 900,
    seats: 30,
    description: "Comfortable AC bus with reclining seats.",
    photo: ""
  },
  {
    id: 2,
    name: "Night Sleeper",
    category: "Sleeper Coach",
    from: "Chennai",
    to: "Madurai",
    time: "21:00",
    price: 1200,
    seats: 24,
    description: "Sleeper coach for overnight journeys.",
    photo: ""
  },
  {
    id: 3,
    name: "City Travels",
    category: "Seater Coach",
    from: "Bangalore",
    to: "Chennai",
    time: "10:00",
    price: 700,
    seats: 35,
    description: "Affordable seater coach.",
    photo: ""
  }
];

function load(key, fallback) {
  try {
```

```
        const value = localStorage.getItem(key);
        return value ? JSON.parse(value) : fallback;
    } catch {
        return fallback;
    }
}

let buses = load("busgoBuses", defaultBuses);
let users = load("busgoUsers", []);
let tickets = load("busgoTickets", []);
let comments = load("busgoComments", []);
let news = load("busgoNews", []);
let currentUser = load("busgoCurrentUser", null);
let adminLoggedIn = false;
let nextBusId = Math.max(0, ...buses.map(b => b.id)) + 1;

function save() {
    localStorage.setItem("busgoBuses", JSON.stringify(buses));
    localStorage.setItem("busgoUsers", JSON.stringify(users));
    localStorage.setItem("busgoTickets", JSON.stringify(tickets));
    localStorage.setItem("busgoComments", JSON.stringify(comments));
    localStorage.setItem("busgoNews", JSON.stringify(news));
    localStorage.setItem("busgoCurrentUser", JSON.stringify(currentUser));
}

function today() {
    const d = new Date();
    return d.getFullYear() + "-" +
        String(d.getMonth() + 1).padStart(2, "0") + "-" +
        String(d.getDate()).padStart(2, "0");
}

document.getElementById("journeyDate").min = today();
document.getElementById("bookDate").min = today();

function esc(value) {
```

```
    return String(value ?? "").replace(/[&<>"']/g, ch => ({
        "&": "&amp;",
        "<": "&lt;",
        ">": "&gt;",
        "": "&quot;",
        "'": "&#39;"
    })[ch]);
}

function occupiedSeats(busId, date) {
    return tickets
        .filter(t => t.busId === busId &&
            t.date === date && t.status === "Confirmed")
        .map(t => t.seat);
}

function fillBusOptions() {
    const selects = [
        document.getElementById("busSelect"),
        document.getElementById("commentBus")
    ];

    selects.forEach(select => {
        const oldValue = select.value;
        select.innerHTML = '<option value="">Choose a bus</option>';

        buses.forEach(bus => {
            const option = document.createElement("option");
            option.value = bus.id;
            option.textContent =
                bus.name + " - " + bus.from + " to " + bus.to;
            select.appendChild(option);
        });

        if ([...select.options].some(o => o.value === oldValue)) {
            select.value = oldValue;
```

```
        }
    });
}

function renderBuses(list = buses) {
    const area = document.getElementById("searchResults");
    area.innerHTML = "";

    if (!list.length) {
        area.textContent = "No buses found for this route and date.";
        return;
    }

    list.forEach(bus => {
        const card = document.createElement("div");
        card.className = "bus-card";

        card.innerHTML =
            (bus.photo
                ? '<img src="' + esc(bus.photo) + '" alt="Bus photo">'
                : '<div class="bus-icon">🚌</div>') +
            "<h3>" + esc(bus.name) + "</h3>" +
            "<p><b>Category:</b> " + esc(bus.category) + "</p>" +
            "<p>" + esc(bus.from) + " → " + esc(bus.to) + "</p>" +
            "<p><b>Departure:</b> " + esc(bus.time) + "</p>" +
            "<p>" + esc(bus.description) + "</p>" +
            '<p class="bus-price">₹' + bus.price + "</p>" +
            "<p>Total seats: " + bus.seats + "</p>";

        const button = document.createElement("button");
        button.textContent = "Select Bus";
        button.onclick = () => {
            document.getElementById("busSelect").value = bus.id;
            document.getElementById("bookDate").value =
                document.getElementById("journeyDate").value || today();
            updateSeatOptions();
```

```
            document.getElementById("booking").scrollIntoView({
                behavior: "smooth"
            });
        };

        card.appendChild(button);
        area.appendChild(card);
    });
}

document.getElementById("searchForm").addEventListener("submit", function(e) {
    e.preventDefault();

    const from = document.getElementById("from").value.trim().toLowerCase();
    const to = document.getElementById("to").value.trim().toLowerCase();
    const date = document.getElementById("journeyDate").value;

    if (date < today()) {
        alert("Please select today or a future date.");
        return;
    }

    const results = buses.filter(bus =>
        bus.from.toLowerCase() === from &&
        bus.to.toLowerCase() === to
    );

    renderBuses(results);

    if (!results.length) {
        document.getElementById("searchResults").textContent =
            "No buses found. Try Chennai to Bangalore, Chennai to Madurai, or Bangalore to
Chennai.";
    }
});
```

```
document.getElementById("busSelect").addEventListener("change", updateSeatOptions);
document.getElementById("bookDate").addEventListener("change", updateSeatOptions);

function updateSeatOptions() {
    const busId = Number(document.getElementById("busSelect").value);
    const date = document.getElementById("bookDate").value;
    const select = document.getElementById("seatSelect");
    const message = document.getElementById("seatMessage");

    select.innerHTML = '<option value="">Select a seat</option>';

    const bus = buses.find(b => b.id === busId);

    if (!bus || !date) {
        message.textContent = "Choose a bus and date to see available seats.";
        return;
    }

    const occupied = occupiedSeats(busId, date);
    let count = 0;

    for (let i = 1; i <= bus.seats; i++) {
        if (!occupied.includes(i)) {
            const option = document.createElement("option");
            option.value = i;
            option.textContent = "Seat " + i;
            select.appendChild(option);
            count++;
        }
    }

    message.textContent = count + " seat(s) available.";
}

document.getElementById("registerForm").addEventListener("submit", function(e) {
    e.preventDefault();
```

```
  const name = document.getElementById("regName").value.trim();
  const email = document.getElementById("regEmail").value.trim().toLowerCase();
  const password = document.getElementById("regPassword").value;

  if (users.some(u => u.email === email)) {
    document.getElementById("accountMessage").textContent =
      "This email is already registered.";
    return;
  }

  users.push({ name, email, password });
  currentUser = { name, email };
  save();
  renderHistory();

  document.getElementById("accountMessage").textContent =
    "Registration successful. You are logged in.";
  this.reset();
});

document.getElementById("loginForm").addEventListener("submit", function(e) {
  e.preventDefault();

  const email = document.getElementById("loginEmail").value.trim().toLowerCase();
  const password = document.getElementById("loginPassword").value;

  const user = users.find(u => u.email === email && u.password === password);

  if (!user) {
    document.getElementById("accountMessage").textContent =
      "Invalid login. Please register or check your details.";
    return;
  }

  currentUser = { name: user.name, email: user.email };
```

```
    save();
    renderHistory();

    document.getElementById("accountMessage").textContent =
        "Welcome, " + user.name + "!";
    this.reset();
});

function logout() {
    currentUser = null;
    save();
    renderHistory();
    document.getElementById("accountMessage").textContent = "Logged out.";
}

function forgotPassword() {
    const email = prompt("Enter your registered email address:");
    if (!email) return;

    const user = users.find(u =>
        u.email === email.trim().toLowerCase()
    );

    alert(user
        ? "Demo: password reset requires a real email service in a deployed system."
        : "No account found for that email.");
}

document.getElementById("bookingForm").addEventListener("submit", function(e) {
    e.preventDefault();

    if (!currentUser) {
        document.getElementById("bookingMessage").textContent =
            "Please register or log in before booking.";
        return;
    }
```

```
const busId = Number(document.getElementById("busSelect").value);
const bus = buses.find(b => b.id === busId);
const date = document.getElementById("bookDate").value;
const seat = Number(document.getElementById("seatSelect").value);
const email = document.getElementById("passengerEmail").value.trim().toLowerCase();

if (!bus || !date || date < today() || !seat) {
    document.getElementById("bookingMessage").textContent =
        "Please select a bus, a valid journey date and an available seat.";
    return;
}

if (email !== currentUser.email) {
    document.getElementById("bookingMessage").textContent =
        "Use the email address of your logged-in account.";
    return;
}

if (occupiedSeats(busId, date).includes(seat)) {
    document.getElementById("bookingMessage").textContent =
        "That seat is already booked. Please select another seat.";
    updateSeatOptions();
    return;
}

const ticket = {
    id: Date.now(),
    name: document.getElementById("passengerName").value.trim(),
    email,
    age: Number(document.getElementById("passengerAge").value),
    busId,
    busName: bus.name,
    from: bus.from,
    to: bus.to,
    date,
```

```
        seat,
        price: bus.price,
        status: "Confirmed"
    };

    tickets.push(ticket);
    save();
    renderHistory();
    renderAdmin();

    document.getElementById("bookingMessage").textContent =
        "Ticket booked successfully! Ticket ID: " + ticket.id +
        ". Amount: ₹" + ticket.price +
        ". Demo booking only; no payment was processed.";

    this.reset();
    updateSeatOptions();
});

function renderHistory() {
    const area = document.getElementById("ticketHistory");
    area.innerHTML = "";

    if (!currentUser) {
        area.textContent = "Please log in to view your tickets.";
        return;
    }

    const mine = tickets.filter(t => t.email === currentUser.email);

    if (!mine.length) {
        area.textContent = "You have no bookings yet.";
        return;
    }

    mine.forEach(t => {
```

```
        const card = document.createElement("div");

        card.innerHTML =
            "<h3>🎫 Ticket #" + t.id + "</h3>" +
            "<p>Bus: " + esc(t.busName) + "</p>" +
            "<p>Route: " + esc(t.from) + " → " + esc(t.to) + "</p>" +
            "<p>Date: " + esc(t.date) + "</p>" +
            "<p>Seat: " + t.seat + "</p>" +
            "<p>Passenger: " + esc(t.name) + "</p>" +
            "<p>Fare: ₹" + t.price + "</p>" +
            "<p>Status: " + esc(t.status) + "</p>";

        if (t.status === "Confirmed") {
            const cancel = document.createElement("button");
            cancel.textContent = "Cancel Ticket";
            cancel.onclick = () => cancelTicket(t.id);
            card.appendChild(cancel);
        }

        area.appendChild(card);
    });
}

function cancelTicket(id) {
    const ticket = tickets.find(t => t.id === id);

    if (!ticket || ticket.status !== "Confirmed") return;

    if (confirm("Are you sure you want to cancel this ticket?")) {
        ticket.status = "Cancelled";
        save();
        renderHistory();
        renderAdmin();
        updateSeatOptions();
    }
}
```

```
document.getElementById("commentForm").addEventListener("submit", function(e) {
    e.preventDefault();

    const busId = Number(document.getElementById("commentBus").value);
    const bus = buses.find(b => b.id === busId);

    if (!bus) return;

    comments.push({
        name: document.getElementById("commentName").value.trim(),
        busName: bus.name,
        text: document.getElementById("commentText").value.trim()
    });

    save();
    renderComments();
    this.reset();
});

function renderComments() {
    const area = document.getElementById("commentList");
    area.innerHTML = "";

    comments.forEach(c => {
        const card = document.createElement("div");
        card.innerHTML =
            "<h3>" + esc(c.busName) + "</h3>" +
            "<p><b>" + esc(c.name) + "</b></p>" +
            "<p>" + esc(c.text) + "</p>";
        area.appendChild(card);
    });
}

document.getElementById("adminLoginForm").addEventListener("submit", function(e) {
    e.preventDefault();
```

```
    const email = document.getElementById("adminEmail").value.trim();
    const password = document.getElementById("adminPassword").value;

    if (email === "admin@busgo.com" && password === "admin123") {
        adminLoggedIn = true;
        document.getElementById("adminPanel").hidden = false;
        renderAdmin();
        document.getElementById("accountMessage").textContent =
            "Administrator login successful.";
    } else {
        alert("Invalid administrator credentials.");
    }
});

document.getElementById("busForm").addEventListener("submit", function(e) {
    e.preventDefault();

    if (!adminLoggedIn) return;

    const bus = {
        id: nextBusId++,
        name: document.getElementById("busName").value.trim(),
        category: document.getElementById("busCategory").value,
        from: document.getElementById("busFrom").value.trim(),
        to: document.getElementById("busTo").value.trim(),
        time: document.getElementById("busTime").value,
        price: Number(document.getElementById("busPrice").value),
        seats: Number(document.getElementById("busSeats").value),
        photo: document.getElementById("busPhoto").value.trim(),
        description: document.getElementById("busDescription").value.trim()
    };

    buses.push(bus);
    save();
    fillBusOptions();
```

```
    renderBuses();
    renderAdmin();
    this.reset();

    alert("Bus added successfully.");
});

function renderAdmin() {
    if (!adminLoggedIn) return;

    const busArea = document.getElementById("adminBusList");
    busArea.innerHTML = "";

    buses.forEach(bus => {
        const card = document.createElement("div");
        card.innerHTML =
            "<h3>" + esc(bus.name) + "</h3>" +
            "<p>" + esc(bus.from) + " → " + esc(bus.to) + "</p>" +
            "<p>Category: " + esc(bus.category) + "</p>" +
            "<p>Price: ₹" + bus.price + " | Seats: " + bus.seats + "</p>";

        const edit = document.createElement("button");
        edit.textContent = "Update Price / Seats";
        edit.onclick = () => {
            const price = prompt("New ticket price:", bus.price);
            if (price === null) return;

            const seats = prompt("New total seats:", bus.seats);
            if (seats === null) return;

            if (Number(price) <= 0 ||
                !Number.isInteger(Number(seats)) ||
                Number(seats) < 1 ||
                Number(seats) > 60 ||
                Number(seats) < occupiedSeats(bus.id, today()).length) {
                alert("Enter valid price and seat quantity.");
```

```
            return;
        }

        bus.price = Number(price);
        bus.seats = Number(seats);
        tickets.filter(t => t.busId === bus.id && t.status === "Confirmed")
            .forEach(t => t.price = bus.price);

        save();
        fillBusOptions();
        renderBuses();
        renderAdmin();
    };

    const remove = document.createElement("button");
    remove.textContent = "Delete Bus";
    remove.onclick = () => {
        if (tickets.some(t => t.busId === bus.id && t.status === "Confirmed")) {
            alert("Cancel the bus's confirmed tickets before deleting it.");
            return;
        }

        if (confirm("Delete this bus?")) {
            buses = buses.filter(b => b.id !== bus.id);
            save();
            fillBusOptions();
            renderBuses();
            renderAdmin();
        }
    };

    card.appendChild(edit);
    card.appendChild(remove);
    busArea.appendChild(card);
});
```

```
const userArea = document.getElementById("adminUserList");
userArea.innerHTML = "";

users.forEach(user => {
    const card = document.createElement("div");
    card.innerHTML =
        "<p>Name: " + esc(user.name) + "</p>" +
        "<p>Email: " + esc(user.email) + "</p>";
    userArea.appendChild(card);
});

const ticketArea = document.getElementById("adminTicketList");
ticketArea.innerHTML = "";

tickets.forEach(t => {
    const card = document.createElement("div");
    card.innerHTML =
        "<p>Ticket #" + t.id + " | " + esc(t.name) + "</p>" +
        "<p>" + esc(t.busName) + " | " + esc(t.date) + "</p>" +
        "<p>Status: " + esc(t.status) + "</p>";

    if (t.status === "Confirmed") {
        const cancel = document.createElement("button");
        cancel.textContent = "Cancel Ticket";
        cancel.onclick = () => {
            t.status = "Cancelled";
            save();
            renderAdmin();
            renderHistory();
            updateSeatOptions();
        };
        card.appendChild(cancel);
    }

    ticketArea.appendChild(card);
});
```

```
    const newsArea = document.getElementById("newsOutput");
    newsArea.innerHTML = "";

    news.forEach(item => {
        const div = document.createElement("div");
        div.textContent = item;
        newsArea.appendChild(div);
    });
}

function generateReport() {
    if (!adminLoggedIn) return;

    const confirmed = tickets.filter(t => t.status === "Confirmed");
    const cancelled = tickets.filter(t => t.status === "Cancelled");
    const revenue = confirmed.reduce((sum, t) => sum + t.price, 0);

    document.getElementById("reportOutput").innerHTML =
        "<h3>Booking Report</h3>" +
        "<p>Total buses: " + buses.length + "</p>" +
        "<p>Registered users: " + users.length + "</p>" +
        "<p>Confirmed tickets: " + confirmed.length + "</p>" +
        "<p>Cancelled tickets: " + cancelled.length + "</p>" +
        "<p>Confirmed ticket value: ₹" + revenue + "</p>";

    const csv = [
        ["Ticket ID", "Passenger", "Email", "Bus", "Date", "Seat", "Fare", "Status"],
        ...tickets.map(t => [
            t.id, t.name, t.email, t.busName, t.date, t.seat, t.price, t.status
        ])
    ].map(row => row.map(value =>
        '"' + String(value).replace(/"/g, '""') + '"'
    ).join(",")).join("\r\n");

    const blob = new Blob([csv], { type: "text/csv;charset=utf-8;" });
```

```
    const url = URL.createObjectURL(blob);
    const link = document.createElement("a");
    link.href = url;
    link.download = "BusGo_Report.csv";
    link.click();
    URL.revokeObjectURL(url);
}

function saveNews() {
    if (!adminLoggedIn) return;

    const text = document.getElementById("newsText").value.trim();
    if (!text) return;

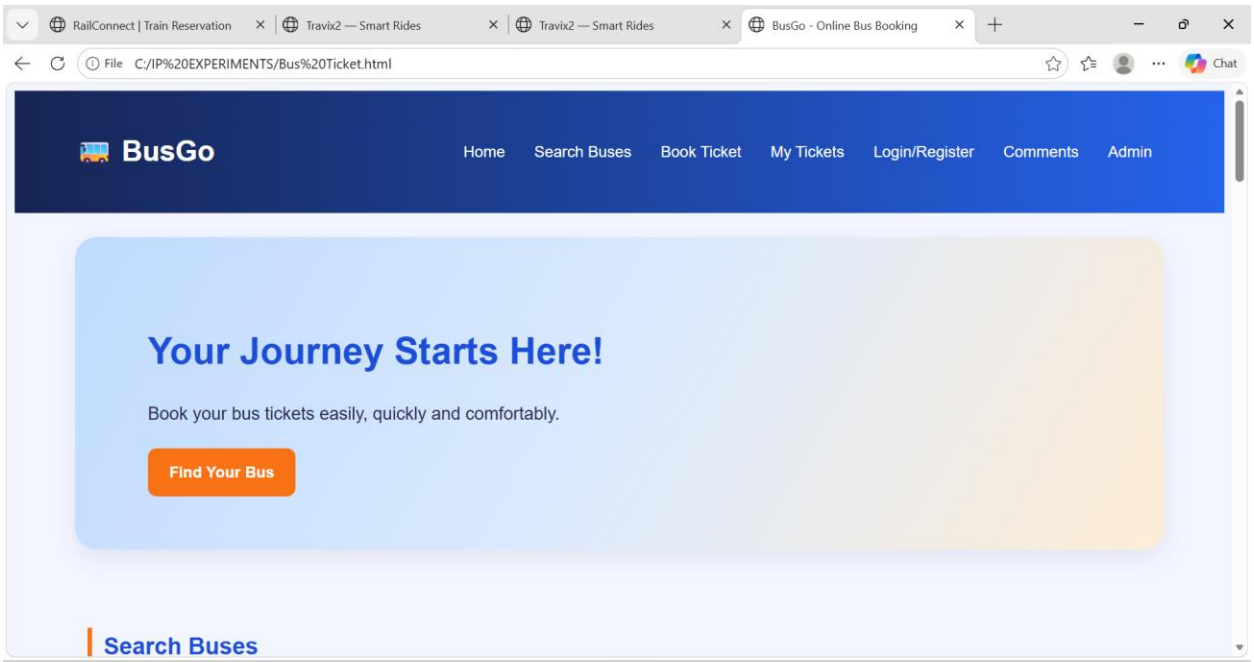
    news.push(text);
    save();
    renderAdmin();
    document.getElementById("newsText").value = "";
}

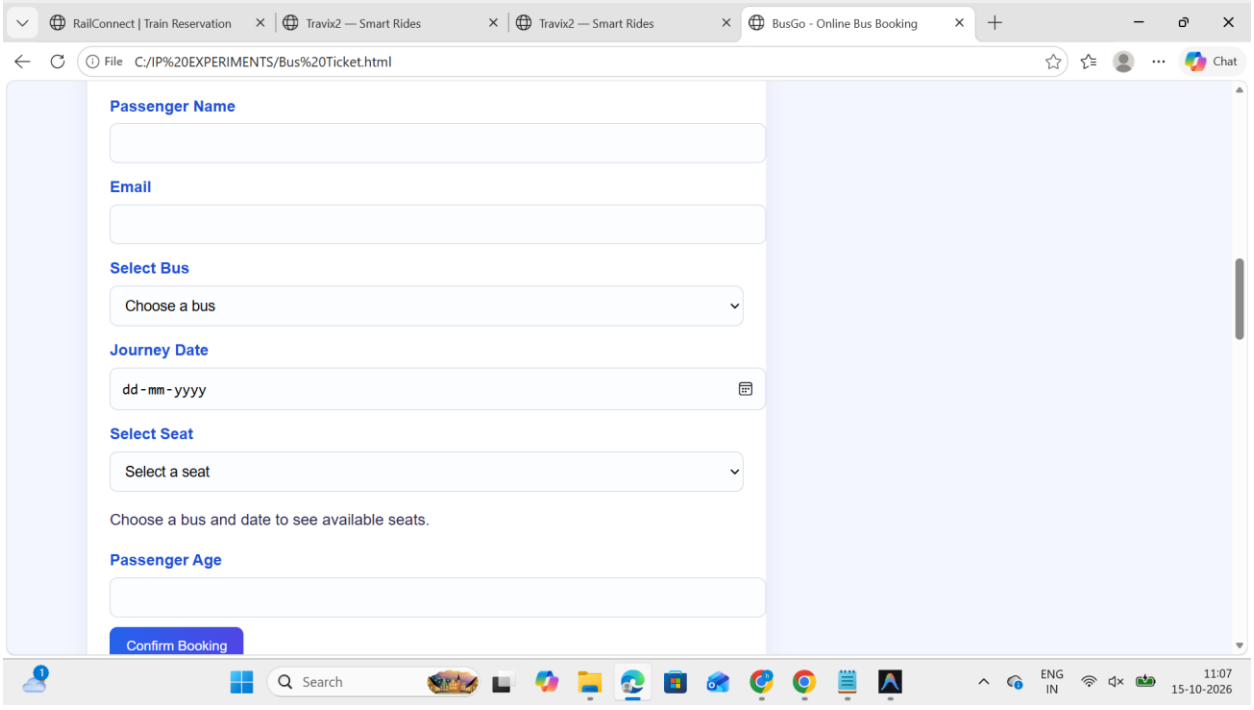
fillBusOptions();
renderBuses();
renderHistory();
renderComments();
```

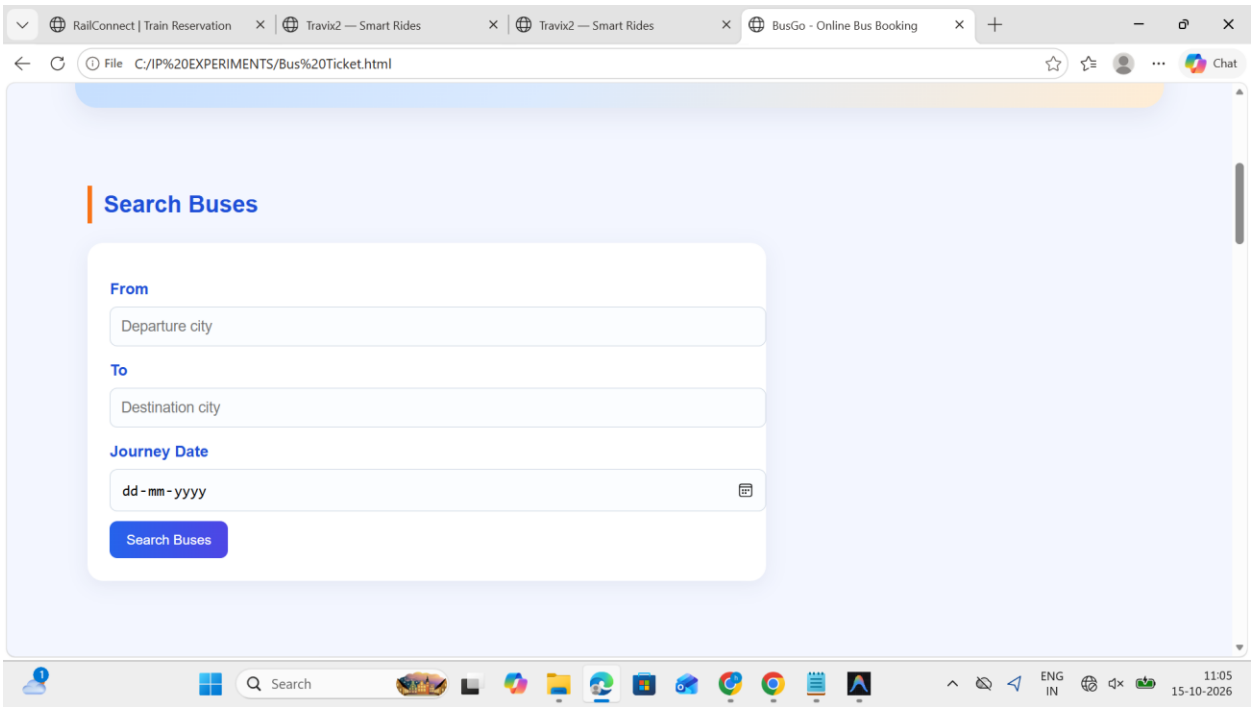
Result

The bus ticket booking system was developed and tested successfully. Users could search buses, book and cancel tickets and post comments, and the admin could manage bus categories, bus details, users and reports.

OUTPUT:


RailConnect | Train Reservation
Travix2 — Smart Rides
Travix2 — Smart Rides
BusGo - Online Bus Booking
File C:/IP%20EXPERIMENTS/Bus%20Ticket.html
Chat
BusGo
Home
Search Buses
Book Ticket
My Tickets
Login/Register
Comments
Admin
Your Journey Starts Here!
Book your bus tickets easily, quickly and comfortably.
Find Your Bus
Search Buses
Search
ENG
IN
11:06
15-10-2026


RailConnect | Train Reservation
Travix2 — Smart Rides
Travix2 — Smart Rides
BusGo - Online Bus Booking
File C:/IP%20EXPERIMENTS/Bus%20Ticket.html
Chat
Passenger Name
Email
Select Bus
Choose a bus
Journey Date
dd-mm-yyyy
Select Seat
Select a seat
Choose a bus and date to see available seats.
Passenger Age
Confirm Booking
Search
ENG
IN
11:07
15-10-2026


RailConnect | Train Reservation
Travix2 — Smart Rides
Travix2 — Smart Rides
BusGo - Online Bus Booking
File C:/IP%20EXPERIMENTS/Bus%20Ticket.html
Chat
Search Buses
From
Departure city
To
Destination city
Journey Date
dd-mm-yyyy
Search Buses
Search
ENG
IN
11:05
15-10-2026